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Java SE 27 & JDK 27

Interface DataInput

All Known Subinterfaces:

`ImageInputStream`, `ImageOutputStream`, `ObjectInput`

All Known Implementing Classes:

`DataInputStream`, `FileCacheImageInputStream`, `FileCacheImageOutputStream`, `FileImageInputStream`, `FileImageOutputStream`, `ImageInputStreamImpl`, `ImageOutputStreamImpl`, `MemoryCacheImageInputStream`, `MemoryCacheImageOutputStream`, `ObjectInputStream`, `RandomAccessFile`

`public interface` **`DataInput`**

The `DataInput` interface provides for reading bytes from a binary stream and reconstructing from them data in any of the Java primitive types. There is also a facility for reconstructing a `String` from data in modified UTF-8 format.

It is generally true of all the reading routines in this interface that if end of file is reached before the desired number of bytes has been read, an `EOFException` (which is a kind of `IOException`) is thrown. If any byte cannot be read for any reason other than end of file, an `IOException` other than `EOFException` is thrown. In particular, an `IOException` may be thrown if the input stream has been closed.

Modified UTF-8

Implementations of the DataInput and DataOutput interfaces represent Unicode strings in a format that is a slight modification of UTF-8. (For information regarding the standard UTF-8 format, see section *3.9 Unicode Encoding Forms* of *The Unicode Standard, Version 4.0*)

- Characters in the range `'\u0001'` to `'\u007F'` are represented by a single byte.
- The null character `'\u0000'` and characters in the range `'\u0080'` to `'\u07FF'` are represented by a pair of bytes.
- Characters in the range `'\u0800'` to `'\uFFFF'` are represented by three bytes.

Encoding of UTF-8 values

Value	**Byte**	**Bit Values**							
		7	**6**	**5**	**4**	**3**	**2**	**1**	**0**
`\u0001` to `\u007F`	1	0	bits 6-0						

Value	Byte	Bit Values							
		7	6	5	4	3	2	1	0
\u0000, \u0080 to \u07FF	1	1	1	0	bits 10-6				
	2	1	0	bits 5-0					
\u0800 to \uFFFF	1	1	1	1	0	bits 15-12			
	2	1	0	bits 11-6					
	3	1	0	bits 5-0					

The differences between this format and the standard UTF-8 format are the following:

- The null byte `'\u0000'` is encoded in 2-byte format rather than 1-byte, so that the encoded strings never have embedded nulls.
- Only the 1-byte, 2-byte, and 3-byte formats are used.
- Supplementary characters are represented in the form of surrogate pairs.

Since:

1.0

See Also:

`DataInputStream`, `DataOutput`

Method Summary

All Methods | Instance Methods

Modifier and Type	Method	Description
`boolean`	`readBoolean()`	Reads one input byte and returns `true` if that byte is nonzero, `false` if that byte is zero.
`byte`	`readByte()`	Reads and returns one input byte.
`char`	`readChar()`	Reads two input bytes and returns a `char` value.
`double`	`readDouble()`	Reads eight input bytes and returns a `double` value.

`float`	**`readFloat`**`()`	Reads four input bytes and returns a `float` value.
`void`	**`readFully`**`(byte[] b)`	Reads some bytes from an input stream and stores them into the buffer array `b`.
`void`	**`readFully`**`(byte[] b, int off, int len)`	Reads `len` bytes from an input stream.
`int`	**`readInt`**`()`	Reads four input bytes and returns an `int` value.
`String`	**`readLine`**`()`	Reads the next line of text from the input stream.
`long`	**`readLong`**`()`	Reads eight input bytes and returns a `long` value.
`short`	**`readShort`**`()`	Reads two input bytes and returns a `short` value.
`int`	**`readUnsignedByte`**`()`	Reads one input byte, zero-extends it to type `int`, and returns the result, which is therefore in the range `0` through `255`.
`int`	**`readUnsignedShort`**`()`	Reads two input bytes and returns an `int` value in the range `0` through `65535`.
`String`	**`readUTF`**`()`	Reads in a string that has been encoded using a modified UTF-8 format.
`int`	**`skipBytes`**`(int n)`	Makes an attempt to skip over `n` bytes of data from the input stream,

	discarding the skipped bytes.

Method Details

readFully

```
void readFully(byte[] b)
        throws IOException
```

Reads some bytes from an input stream and stores them into the buffer array `b`. The number of bytes read is equal to the length of `b`.

This method blocks until one of the following conditions occurs:

- `b.length` bytes of input data are available, in which case a normal return is made.
- End of file is detected, in which case an `EOFException` is thrown.
- An I/O error occurs, in which case an `IOException` other than `EOFException` is thrown.

If `b` is `null`, a `NullPointerException` is thrown. If `b.length` is zero, then no bytes are read. Otherwise, the first byte read is stored into element `b[0]`, the next one into `b[1]`, and so on. If an exception is thrown from this method, then it may be that some but not all bytes of `b` have been updated with data from the input stream.

Parameters:

`b` - the buffer into which the data is read.

Throws:

`NullPointerException` - if `b` is `null`.

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

readFully

```
void readFully(byte[] b,
               int off,
               int len)
        throws IOException
```

Reads `len` bytes from an input stream.

This method blocks until one of the following conditions occurs:

- `len` bytes of input data are available, in which case a normal return is made.
- End of file is detected, in which case an `EOFException` is thrown.
- An I/O error occurs, in which case an `IOException` other than `EOFException` is thrown.

If `b` is `null`, a `NullPointerException` is thrown. If `off` is negative, or `len` is negative, or `off+len` is greater than the length of the array `b`, then an `IndexOutOfBoundsException` is thrown. If `len` is zero, then no bytes are read. Otherwise, the first byte read is stored into element `b[off]`, the next one into `b[off+1]`, and so on. The number of bytes read is, at most, equal to `len`.

Parameters:

`b` - the buffer into which the data is read.

`off` - an int specifying the offset in the data array `b`.

`len` - an int specifying the number of bytes to read.

Throws:

`NullPointerException` - if `b` is `null`.

`IndexOutOfBoundsException` - if `off` is negative, `len` is negative, or `len` is greater than `b.length - off`.

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

skipBytes

```
int skipBytes(int n)
       throws IOException
```

Makes an attempt to skip over `n` bytes of data from the input stream, discarding the skipped bytes. However, it may skip over some smaller number of bytes, possibly zero. This may result from any of a number of conditions; reaching end of file before `n` bytes have been skipped is only one possibility. This method never throws an `EOFException`. The actual number of bytes skipped is returned.

Parameters:

`n` - the number of bytes to be skipped.

Returns:

the number of bytes actually skipped.

Throws:

`IOException` - if an I/O error occurs.

readBoolean

```
boolean readBoolean()
             throws IOException
```

Reads one input byte and returns `true` if that byte is nonzero, `false` if that byte is zero. This method is suitable for reading the byte written by the `writeBoolean` method of interface `DataOutput`.

Returns:

the `boolean` value read.

Throws:

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

readByte

```
byte readByte()
       throws IOException
```

Reads and returns one input byte. The byte is treated as a signed value in the range `-128` through `127`, inclusive. This method is suitable for reading the byte written by the `writeByte` method of interface `DataOutput`.

Returns:

the 8-bit value read.

Throws:

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

readUnsignedByte

```
int readUnsignedByte()
              throws IOException
```

Reads one input byte, zero-extends it to type `int`, and returns the result, which is therefore in the range `0` through `255`. This method is suitable for reading the byte written by the `writeByte` method of interface `DataOutput` if the argument to `writeByte` was intended to be a value in the range `0` through `255`.

Returns:

the unsigned 8-bit value read.

Throws:

`EOFException` - if this stream reaches the end before reading all the bytes.

IOException - if an I/O error occurs.

readShort

```
short readShort()
          throws IOException
```

Reads two input bytes and returns a `short` value. Let `a` be the first byte read and `b` be the second byte. The value returned is:

```
(short)((a << 8) | (b & 0xff))
```

This method is suitable for reading the bytes written by the `writeShort` method of interface `DataOutput`.

Returns:

the 16-bit value read.

Throws:

EOFException - if this stream reaches the end before reading all the bytes.

IOException - if an I/O error occurs.

readUnsignedShort

```
int readUnsignedShort()
                throws IOException
```

Reads two input bytes and returns an `int` value in the range `0` through `65535`. Let `a` be the first byte read and `b` be the second byte. The value returned is:

```
(((a & 0xff) << 8) | (b & 0xff))
```

This method is suitable for reading the bytes written by the `writeShort` method of interface `DataOutput` if the argument to `writeShort` was intended to be a value in the range `0` through `65535`.

Returns:

the unsigned 16-bit value read.

Throws:

EOFException - if this stream reaches the end before reading all the bytes.

IOException - if an I/O error occurs.

readChar

```
char readChar()
        throws IOException
```

Reads two input bytes and returns a `char` value. Let `a` be the first byte read and `b` be the second byte. The value returned is:

```
(char)((a << 8) | (b & 0xff))
```

This method is suitable for reading bytes written by the `writeChar` method of interface `DataOutput`.

Returns:

the `char` value read.

Throws:

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

readInt

```
int readInt()
    throws IOException
```

Reads four input bytes and returns an `int` value. Let `a-d` be the first through fourth bytes read. The value returned is:

```
(((a & 0xff) << 24) | ((b & 0xff) << 16) |
 ((c & 0xff) <<  8) | (d & 0xff))
```

This method is suitable for reading bytes written by the `writeInt` method of interface `DataOutput`.

Returns:

the `int` value read.

Throws:

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

readLong

```
long readLong()
       throws IOException
```

Reads eight input bytes and returns a `long` value. Let `a-h` be the first through eighth bytes read. The value returned is:

```
(((long)(a & 0xff) << 56) |
 ((long)(b & 0xff) << 48) |
```

```
  ((long)(c & 0xff) << 40) |
  ((long)(d & 0xff) << 32) |
  ((long)(e & 0xff) << 24) |
  ((long)(f & 0xff) << 16) |
  ((long)(g & 0xff) <<  8) |
  ((long)(h & 0xff)))
```

This method is suitable for reading bytes written by the `writeLong` method of interface `DataOutput`.

Returns:

the `long` value read.

Throws:

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

readFloat

```
float readFloat()
          throws IOException
```

Reads four input bytes and returns a `float` value. It does this by first constructing an `int` value in exactly the manner of the `readInt` method, then converting this `int` value to a `float` in exactly the manner of the method `Float.intBitsToFloat`. This method is suitable for reading bytes written by the `writeFloat` method of interface `DataOutput`.

Returns:

the `float` value read.

Throws:

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

readDouble

```
double readDouble()
            throws IOException
```

Reads eight input bytes and returns a `double` value. It does this by first constructing a `long` value in exactly the manner of the `readLong` method, then converting this `long` value to a `double` in exactly the manner of the method `Double.longBitsToDouble`. This method is suitable for reading bytes written by the `writeDouble` method of interface `DataOutput`.

Returns:

the `double` value read.

Throws:

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

readLine

```
String readLine()
         throws IOException
```

Reads the next line of text from the input stream. It reads successive bytes, converting each byte separately into a character, until it encounters a line terminator or end of file; the characters read are then returned as a `String`. Note that because this method processes bytes, it does not support input of the full Unicode character set.

If end of file is encountered before even one byte can be read, then `null` is returned. Otherwise, each byte that is read is converted to type `char` by zero-extension. If the character `'\n'` is encountered, it is discarded and reading ceases. If the character `'\r'` is encountered, it is discarded and, if the following byte converts to the character `'\n'`, then that is discarded also; reading then ceases. If end of file is encountered before either of the characters `'\n'` and `'\r'` is encountered, reading ceases. Once reading has ceased, a `String` is returned that contains all the characters read and not discarded, taken in order. Note that every character in this string will have a value less than `\u0100`, that is, `(char)256`.

Returns:

the next line of text from the input stream, or `null` if the end of file is encountered before a byte can be read.

Throws:

`IOException` - if an I/O error occurs.

readUTF

```
String readUTF()
        throws IOException
```

Reads in a string that has been encoded using a modified UTF-8 format. The general contract of `readUTF` is that it reads a representation of a Unicode character string encoded in modified UTF-8 format; this string of characters is then returned as a `String`.

First, two bytes are read and used to construct an unsigned 16-bit integer in exactly the manner of the `readUnsignedShort` method . This integer value is

called the *UTF length* and specifies the number of additional bytes to be read. These bytes are then converted to characters by considering them in groups. The length of each group is computed from the value of the first byte of the group. The byte following a group, if any, is the first byte of the next group.

If the first byte of a group matches the bit pattern `0xxxxxxx` (where `x` means "may be `0` or `1`"), then the group consists of just that byte. The byte is zero-extended to form a character.

If the first byte of a group matches the bit pattern `110xxxxx`, then the group consists of that byte `a` and a second byte `b`. If there is no byte `b` (because byte `a` was the last of the bytes to be read), or if byte `b` does not match the bit pattern `10xxxxxx`, then a `UTFDataFormatException` is thrown. Otherwise, the group is converted to the character:

```
(char)(((a & 0x1F) << 6) | (b & 0x3F))
```

If the first byte of a group matches the bit pattern `1110xxxx`, then the group consists of that byte `a` and two more bytes `b` and `c`. If there is no byte `c` (because byte `a` was one of the last two of the bytes to be read), or either byte `b` or byte `c` does not match the bit pattern `10xxxxxx`, then a `UTFDataFormatException` is thrown. Otherwise, the group is converted to the character:

```
(char)(((a & 0x0F) << 12) | ((b & 0x3F) << 6) | (c & 0x3F))
```

If the first byte of a group matches the pattern `1111xxxx` or the pattern `10xxxxxx`, then a `UTFDataFormatException` is thrown.

If end of file is encountered at any time during this entire process, then an `EOFException` is thrown.

After every group has been converted to a character by this process, the characters are gathered, in the same order in which their corresponding groups were read from the input stream, to form a `String`, which is returned.

The `writeUTF` method of interface `DataOutput` may be used to write data that is suitable for reading by this method.

Returns:

a Unicode string.

Throws:

`EOFException` - if this stream reaches the end before reading all the bytes.

`IOException` - if an I/O error occurs.

`UTFDataFormatException` - if the bytes do not represent a valid modified UTF-8 encoding of a string.

Report a bug or suggest an enhancement
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